

Memory

2025 Semester 1 SOFTENG 370: Operating Systems
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Lecture 13
1.0.0

Paging Revisit

- `int A[100];`
- All instructions are on page 0.
- Each page holds 64 entries (64 integers).
- Array A starts at location 64.

Page Replacement Algorithm

Memory can hold only a limited number of pages.

When memory is full, a page must be swapped out to swap space (where?).

The OS decides which page to replace using a Page Replacement Algorithm.

Goal: Keep the most useful pages in memory to improve performance (reduce the number of page faults).

Optimal Example

Assume our physical memory can only hold 4 pages, and we access the following:

1 2 3 4 1 2 5 1 2 3 4 5 (all of the pages are initially on disk)

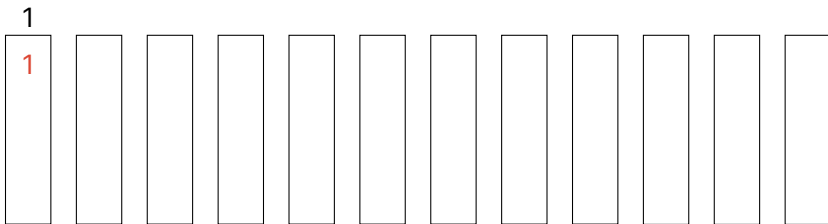
1	2	3	4	1	2	5	1	2	3	4	5
1	1	1	1	1	1	1	1	1	1	4	4
	2	2	2	2	2	2	2	2	2	2	2
		3	3	3	3	3	3	3	3	3	3
			4	4	4	5	5	5	5	5	5

6 page faults

FIFO Example

Assume our physical memory can only hold 4 pages, and we access the following:

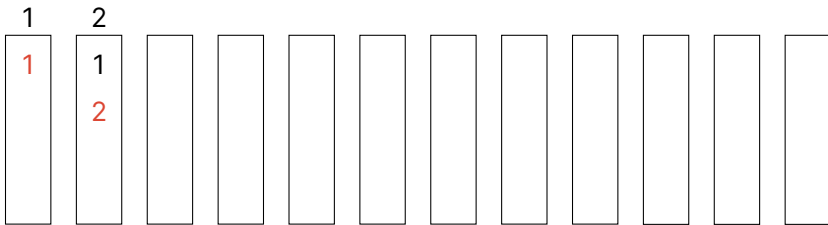
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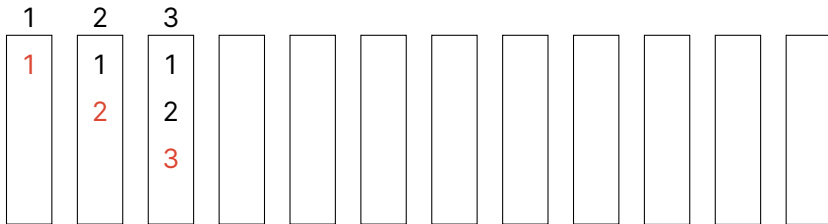
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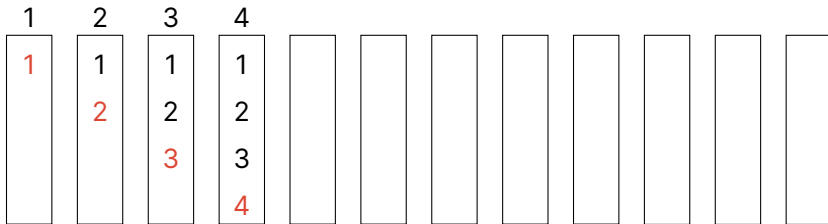
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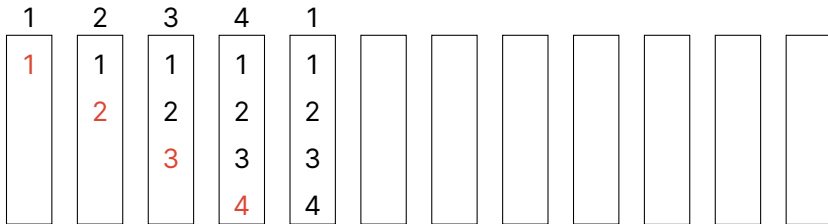
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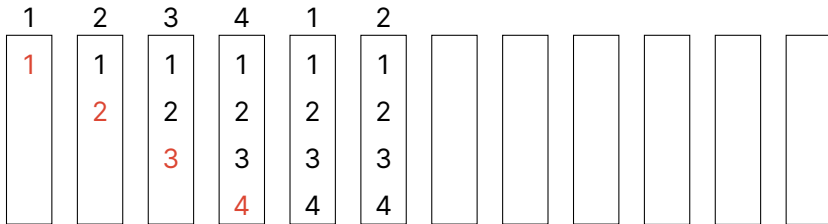
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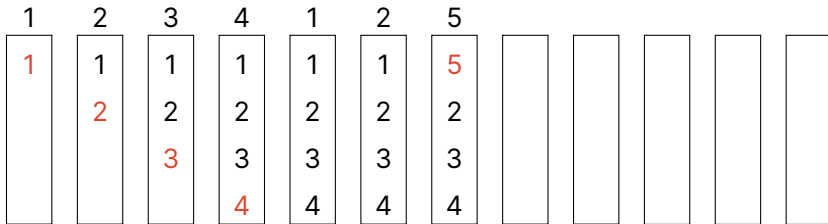
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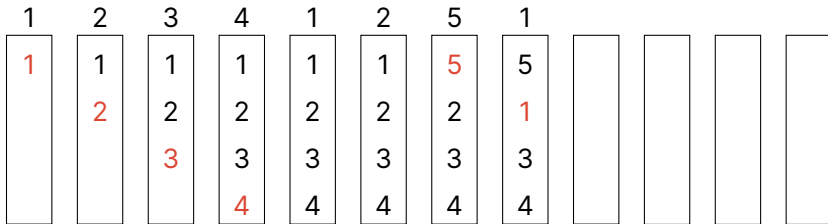
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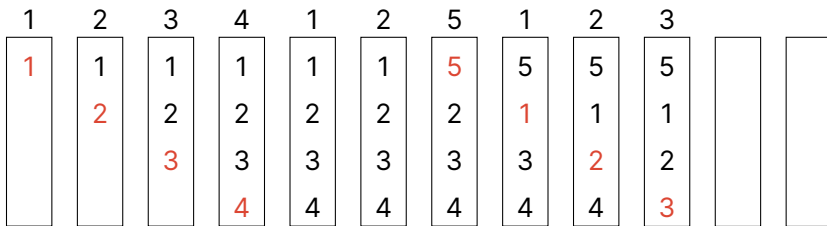
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1	2	3	4	1	2	5	1	2	3	4	
1	1	1	1	1	1	5	5	5	5	4	
	2	2	2	2	2	2	1	1	1	1	
		3	3	3	3	3	3	2	2	2	
			4	4	4	4	4	4	3	3	

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1	1	1	1	1	1	5	5	5	5	4	4
	2	2	2	2	2	2	1	1	1	1	5
		3	3	3	3	3	3	2	2	2	2
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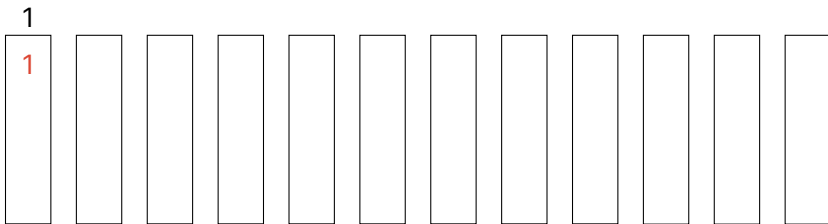
1	2	3	4	1	2	5	1	2	3	4	5
1	1	1	1	1	1	5	5	5	5	4	4
	2	2	2	2	2	2	1	1	1	1	5
		3	3	3	3	3	3	2	2	2	2
			4	4	4	4	4	4	3	3	3

10 page faults

LRU Example (Use FIFO to Break Ties)

Assume our physical memory can only hold 4 pages, and we access the following:

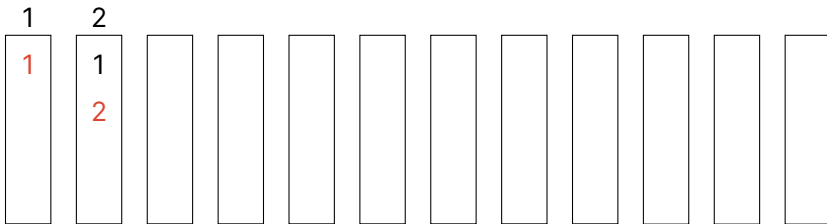
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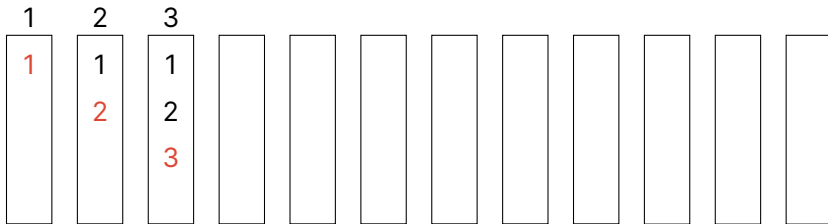
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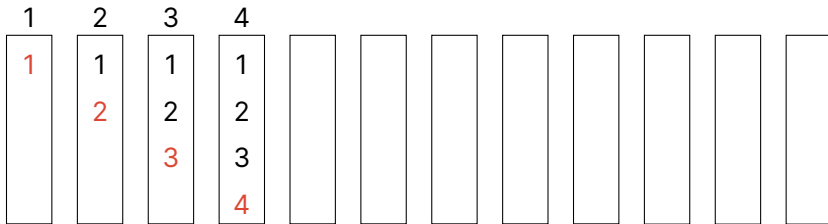
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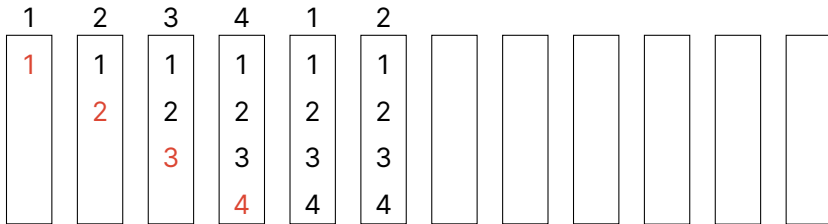
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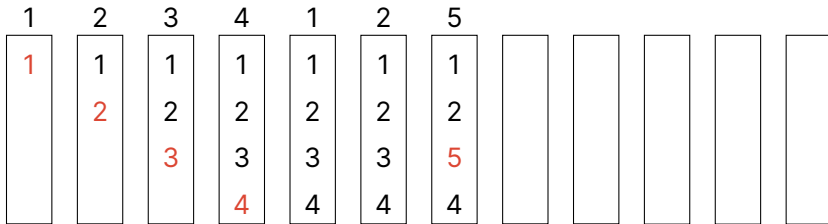
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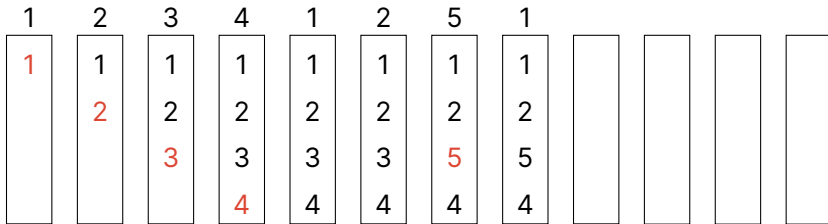
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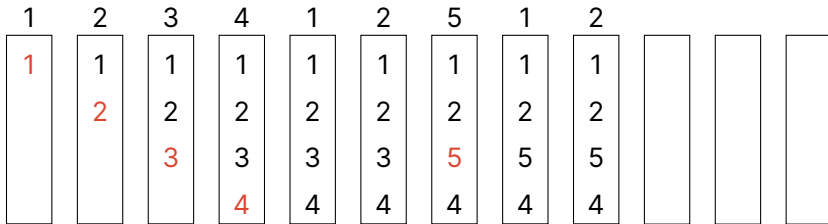
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1	2	3	4	1	2	5	1	2	3		
1	1	1	1	1	1	1	1	1	1		
	2	2	2	2	2	2	2	2	2		
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	2	2	2	2	2	2	2	2	2	2	
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	2	2	2	2	2	2	2	2	2	2	2
		3	3	3	3	5	5	5	5	4	4
			4	4	4	4	4	4	3	3	3

8 page faults

Implementing LRU in Practice Isn't Going to Work

We settle for approximate LRU

LRU is an approximation of the optimal case anyways

There's lots of different tweaks you can do to implement it more efficiently

We'll be looking at the clock algorithm, but there's also:

Least Frequently Used (LFU), 2Q, Adaptive Replacement Cache (ARC)

Clock Algorithm

Data structures:

- Keeps a circular list of pages in memory
- Uses a reference bit for each page in memory (light grey in next slides)
- Has a “hand” (iterator) pointing to the last element examined

Algorithm, to insert a new page:

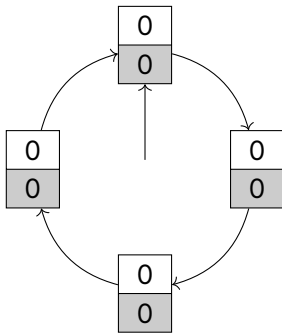
- Check the hand's reference bit, if it's 0 then place the page and advance hand
- If the reference bit is 1, set it to 0, advance the hand, and repeat

For page accesses, set the reference bit to 1

Clock Example (with Diagram)

Assume our physical memory can only hold 4 pages, and we access the following:

1 2 3 4 5 2 3 1 2 3 (all of the pages are initially on disk)

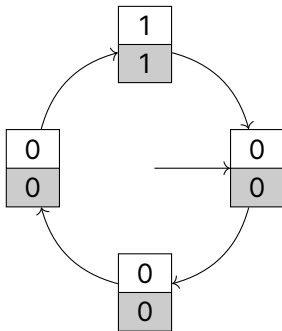


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1 2 3 4 5 2 3 1 2 3 (all of the pages are initially on disk)

1

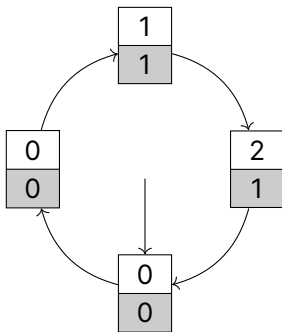


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1 2 3 4 5 2 3 1 2 3 (all of the pages are initially on disk)

1 2

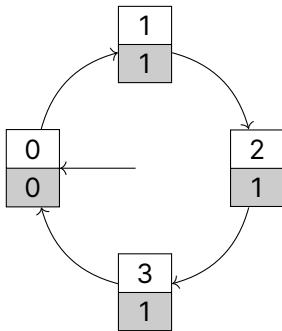


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1 2 3

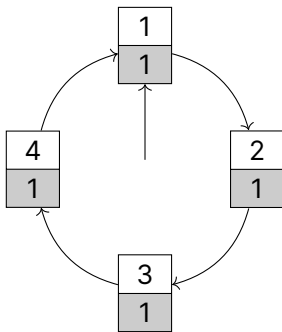


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1 2 3 4

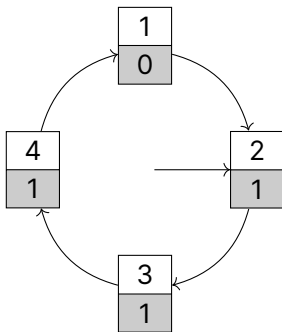


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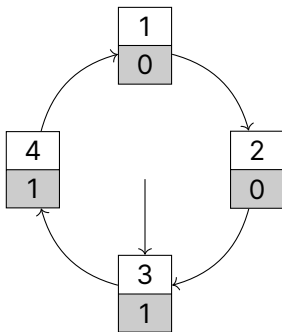


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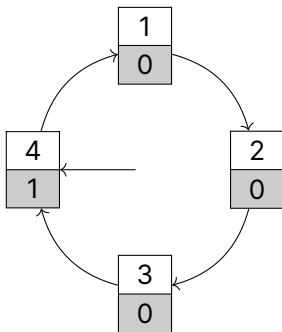


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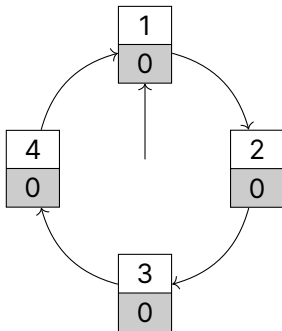


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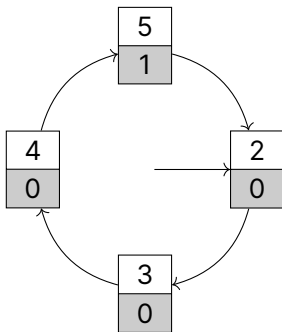


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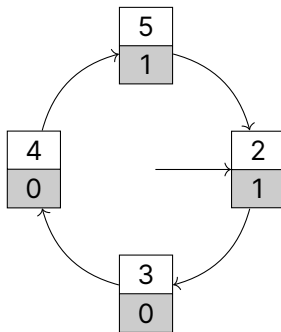


Clock Example (with Diagram)

Assume our physical memory can only hold 4 pages, and we access the following:

1 2 3 4 5 2 3 1 2 3 (all of the pages are initially on disk)

1 2 3 4 5 2

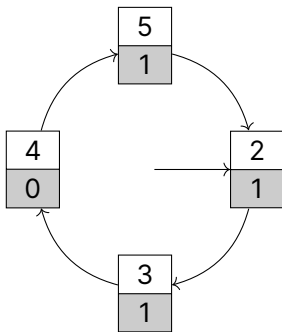


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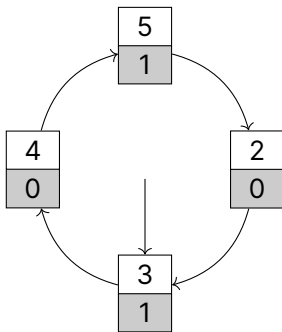


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1 2 3 4 5 2 3 1

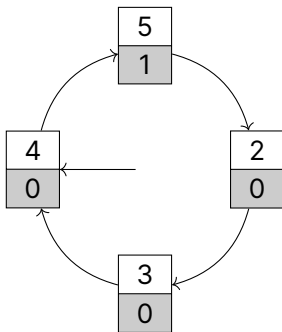


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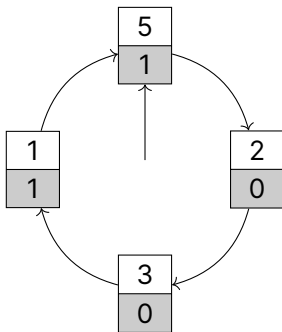


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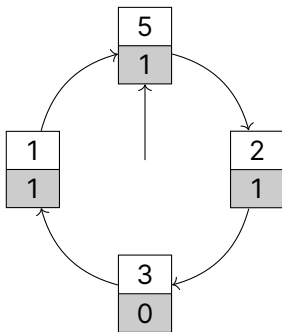


Clock Example (with Diagram)

Assume our physical memory can only hold 4 pages, and we access the following:

1 2 3 4 5 2 3 1 2 3 (all of the pages are initially on disk)

1 2 3 4 5 2 3 1 2

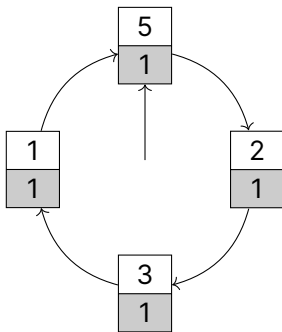


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	2	2	2	2	2	2	2	2	2
		3	3	3	3	3	3	3	3
			4	4	4	4	1	1	1

6 page faults